Journal of Energy Storage 61 (2023) 106772

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Review article

Progress of research on phase change energy storage materials in their thermal conductivity

Shenhui Tan, Xuelai Zhang *

Institute of Cool Thermal Storage Technology, Shanghai Maritime University, Shanghai 201306, China

ARTICLE INFO

Keywords:
Phase change materials
Thermal conductivity
Thermal conductivity mechanism
Thermal conductivity prediction models
Nanomaterials
Metal foams
External fields

ABSTRACT

In recent years, phase change materials (PCM) have become increasingly popular for energy applications due to their unique properties. However, the low thermal conductivity of PCM during phase change can seriously hinder its wide application, so it is crucial to improve the thermal conductivity of PCM. In this paper, the thermal conductivity mechanism of PCM (basic thermal conductivity, phonon thermal conductivity and channel thermal conductivity) and thermal conductivity prediction models (nano-type and foam-metal type) are systematically discussed from a microscopic perspective, providing a theoretical basis and a validation model for the study of thermal conductivity of PCM. In addition, we review three specific approaches to improve PCM from a macroscopic perspective: addition of nanoparticles, embedded foam structures, and addition of external fields. Finally, it is concluded that the specific approaches to improve the thermal conductivity of PCM all proceed to enhance the different thermal conductivity mechanisms of PCM, and future research directions to improve the thermal conductivity of PCM are proposed in response to this conclusion.

1. Introduction

At the 75th session of the United Nations General Assembly, the Chinese government aims to reach peak carbon emissions by 2030 and strives to achieve carbon neutrality by 2060. The conversion and optimization of the energy mix under this goal have become an indispensable part [1]. At present, new energy sources mainly include solar energy, wind energy, tidal energy, etc. However, these energy sources are affected by environmental and time factors with intermittency and randomness, so energy storage becomes especially important and becomes a key technology to solve the energy crisis [2]. The thermal energy storage methods can be classified as sensible heat storage (SHS) [3], latent heat storage (LHS) [4] and thermochemical storage [5], where PCM absorbs and releases heat as latent heat during the phase change. Phase change energy storage materials can solve the uneven distribution of energy in space and time on the one hand, on the other hand, phase change energy storage materials have the advantages of temperature stability, high energy storage density and large energy storage capacity, which makes phase change energy storage materials can be applied to various fields. For example, in the aerospace sector [6], in the building energy-saving field [7], in industrial waste heat utilization [8], and in the apparel industry field [9].

Phase change materials are substances that change the state of matter at constant temperature and can provide latent heat, which can be divided into organic phase change materials, inorganic phase change materials and composite phase change materials, as shown in Fig. 1. Organic PCM has the advantages of high latent heat, wide phase change temperature, non-toxic, non-corrosive and cheap, but also has the disadvantages of low thermal conductivity and easy leakage during phase change [10]. Organic PCM is mainly divided into paraffin and non-paraffin waxes, while non-paraffin waxes are generally fatty acids, alcohols, esters, etc. [11–14] Compared with organic PCM, inorganic PCM has greater thermal conductivity and heat storage, but severe subcooling and phase separation can occur during the phase change process [15]. It mainly has crystalline hydrated salts, molten salts, metal alloys, etc. [16–18] Composite PCMs are generally obtained by compounding two PCMs, adding nanoparticles, and embedding foam structures, as detailed in Part 3 of this paper.

1.1. Research on the thermal conductivity of PCM in recent years

Thermal conductivity is a key parameter for phase change energy storage systems to measure how fast or slow the energy is transferred. Many researchers in China and abroad have done a lot of work on improving the thermal conductivity of phase change materials. Laghari

* Corresponding author.
E-mail address: xlzhang@shmtu.edu.cn (X. Zhang).

https://doi.org/10.1016/j.est.2023.106772
Received 27 October 2022; Received in revised form 15 December 2022; Accepted 27 January 2023
Available online 8 February 2023

Nomenclature

a_1	geometrical parameter controlling cross-sectional shape of ligament, [−]
k	thermal conductivity, [W/(m.K)]
H	height of nanorod, [nm]
R	thermal resistance on a flux basis, [m^2KW^{-1}]
T	temperature, [K]
L	node-to-node length, [m]
C_p	specific heat, [J/kg.K]
a	aspect ratio of Nanoparticle,[−]
e	dimensionless cubic node length,[−]
g	dimensionless foam ligament radius,[−]
t	thickness of the interfacial layer, [nm]
d	diameter,[m]
r	radius of nanorod, [nm]
x,z	x-axis, z-axis,[−]

Greek letters

ϕ	volume fraction of nanoparticles,[%]
ρ	density, [kg/m^3]
δ	The metal foam porosity,[−]
φ	solidity of ligament,[−]
θ	imensionless parameter,[−]

Subscripts

A	unit cell subsection
B	unit cell subsection
C	unit cell subsection
D	unit cell subsection
eff	effective
f	basefluid
s	solid-phase
p	particle (rod)

Abbreviations

PCM	Phase change materials
CBM	Carbon based nanomaterials
CBNP	Carbon black nanoparticles
CNT	Carbon nanotubes
SWCNT	Single-walled carbon nanotubes
MWCNT	Multi-walled carbon nanotubes
GO	Graphene oxide
HTC	Hydrothermal carbon
PEG	Polyethylene glycol
EG	Expanded graphite
GF	Graphite foam
rGOS	Graphene sponge
MP-LA/PAN	Methyl palmitate-lauric acid/acrylonitrile composite phase change fibres

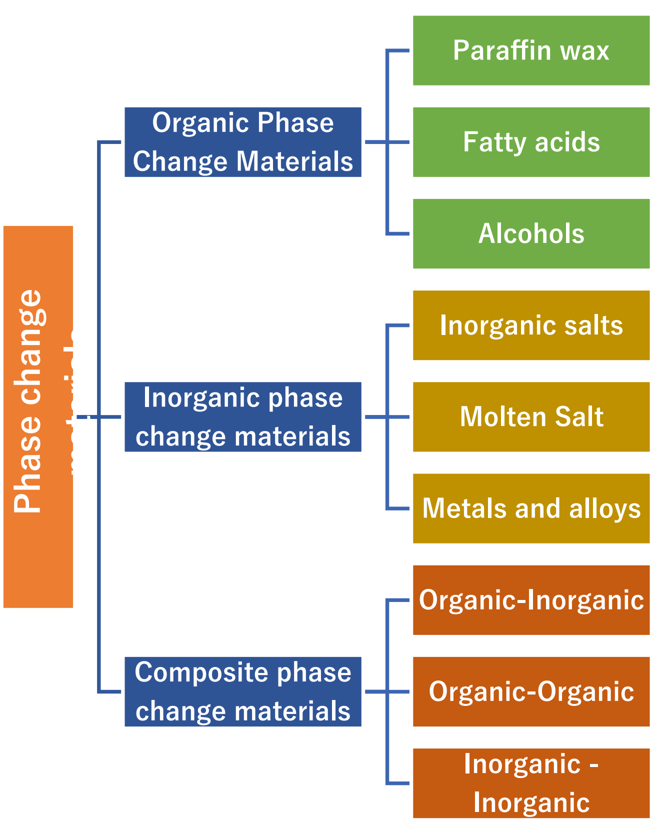

Fig. 1. Schematic diagram of the classification of phase change materials.

et al. [19] improved the thermal conductivity with organic phase change material paraffin wax (PW) as well as its latent heat value by adding titanium dioxide - graphene (TiO_2:*Gr*) binary composites with sodium dodecyl benzene sulfonate (SDBS) as surfactant. Zhou Sunxi et al. [20] adsorbed a binary eutectic mixture of octanoic acid-myristic acid (OA-MA) onto graphite with porous structure, which significantly improved the thermal conductivity of OA-MA. Bao et al. [21] used scale graphite to reduce the subcooling of $CaCl_2{\cdot}6H_2O$ and also used a highly absorbent polymer as a thickener to improve the resistance to dissociation of liquid $CaCl_2{\cdot}6H_2O$, and scale graphite also improved the thermal conductivity of PCM. Hirschey et al. [22] used the extended one-pot synthesis process to prepare salt hydrate-graphite composite materials, and used three kinds of graphite, including expanded graphite (EG), ground expanded graphite (MG) and graphene nanosheet (GnP), to improve the thermal conductivity of sodium sulfate decahydrate (SSD).

In recent years, since PCM can provide heat absorption effect to reduce the operating temperature of the battery, the PCM-based thermal management system for cooling battery has gradually become a hot research topic [23]. However, the vulnerability to leakage, low thermal conductivity and rigidity of PCM limit its application in battery thermal management. To address its low thermal conductivity aspect, Xiangwei Lin et al. [24] used styrene-ethylene-propylene-styrene as a support material to prevent PCM leakage, while expanded graphite was used to improve its thermal conductivity. And Chuyuan *Ma* et al. [25] went to enhance the thermal conductivity of PCM by adding two types of materials, expanded graphite and copper nanoparticles. Also, Yanan Wang et al. [26] found that increasing the thickness of PCM cells can improve the thermal management performance in a passive low-cost battery thermal management system consisting of a pouch Li-ion battery and a phase change material cell. However, exceeding the limit of 6 mm affects the thermal management performance.

1.2. Contribution of this study

Thermal conductivity is one of the most important material properties that affect the performance index of PCM in energy storage systems. In short, the thermal conductivity of PCM directly affects the heat transfer rate during the storage and release of thermal energy, so the efficiency of the energy storage system will be influenced by its thermal conductivity. Based on the importance of phase change energy storage materials in the energy field and the key role of their thermal conductivity parameters. This paper reviews the research progress of PCM in its thermal conductivity from both microscopic and macroscopic perspectives. The contributions of this study are as follows:

- This paper provides a comprehensive summary and classification of the existing thermal conductivity mechanisms and thermal conductivity prediction models for PCM, and provides a theoretical basis for the subsequent development of its novel optimization approach.
- According to the existing studies on the thermal conductivity mechanism of PCM, there is a deficiency. Most of the studies are proposed for solid-liquid PCM, and there are fewer studies on the thermal conductivity mechanisms of both solid-gas and liquid-gas types. This paper provides an idea for the study of solid-gas and liquid-gas thermal conductivity mechanisms in response to the above limitations, as detailed in the outlook (1) in Section 4 of this paper.
- By analyzing the existing PCM thermal conductivity prediction models, it is found that they can be broadly classified into nanotypes and foam-metal types. However, the accuracy of the models needs to be improved, and fewer types of thermal conductivity prediction models other than nanotypes and foam-metal types have been developed, leading to a limited range of applications. This paper provides specific paths to solve the above problems in the outlook(2)(3)(4) in Section 4.
- The existing specific ways to improve the thermal conductivity of PCM are reviewed and can be divided into three types: addition of nanoparticles, embedded foam structures, and addition of external fields. However, most of the existing studies use a single enhancement modality, and fewer studies use more than two modalities to act together on PCM. Therefore, we can combine various methods to improve the thermal conductivity of PCM, and explore its action law to achieve further optimization.

1.3. Outline of this article

The rest of this paper is organized as follows. In Section 2, three PCM thermal conductivity mechanisms and two major thermal conductivity prediction models are analyzed and summarized in detail. In Section 3, three specific methods to improve the thermal conductivity of PCM are reviewed. Finally, in Section 4, its internal mechanism and specific methods for improvement are summarized, along with an outlook on them.

2. Phase change material thermal conductivity mechanism and thermal conductivity prediction model

2.1. Thermal conductivity mechanism of phase change materials

2.1.1. Basic heat transfer mechanism

The basic heat transfer mechanism of phase change materials is mainly heat conduction and convection heat transfer. When the physical state of the phase change material is a solid, the heat transfer on the insulating solid is caused by the thermal vibration of the solid lattice, i.e., the phonon conduction mechanism as described in Section 2.1.2. In contrast, heat conduction on metals and other non-insulated conductors is caused by thermal vibrations of the lattice and free electron movement, i.e. phonon and electron conduction [27]. When the phase change material is in liquid state, the thermal conductivity of the phase change material is convective heat transfer. However, since we know that solid-liquid coexistence is inevitable in the phase change process of phase change materials, it is very important to analyze which is dominant in the process of phase transition, heat conduction or convection heat transfer.

Usually, solid-liquid phase change materials store energy in a solidification process and release energy in a melting process. Domestic and foreign studies for the analysis of the thermal conductivity mechanism during melting and solidification. Firstly, R. Anish et al. [28] found that natural convection of erythritol material during melting had a significant effect on the device by studying the phase transition of erythritol in a horizontal shell multi-pin tube device, and that heat conduction dominated during solidification without subcooling. However, Shiquan He et al. [29] also studied the heat transfer characteristics of organic PCM erythritol phase transition, but clearly subdivided the melting process into three stages, namely, channel formation stage (heat conduction), melting stage (dominated by convection heat transfer) and final stage (storage by sensible heat). There are two stages of latent heat storage: channel formation and melting. This means that the convective heat exchange mechanism gradually increases with the time of the PCM melting process, while the solidification process is the opposite [30].

At the same time, in order to ensure the efficiency of phase change material storage and discharge, it is necessary to reduce the melting/solidification time of PCM from heat conduction and thermal convection mechanism. In the melting process of phase change materials, convective heat transfer is the main heat transfer mechanism. Increasing external factors such as external temperature, external air flow rate and panel tilt angle can influence the promotion of natural convection to reduce melting time and provide energy release efficiency [31]. Wei Cui et al. [32] also enhanced convective heat transfer by using ultrasonic waves on phase change materials to produce a cavitation effect thereby increasing the flow rate of the liquid during the melting process of the phase change material. Xuan Zhang et al. [33] used gallium as the premise of filling phase change material to increase the heat transfer by adding the number of fins found that there is a competing mechanism between thermal conductivity and convective heat transfer during the melting and solidification of the phase change material. As the number of fins increases, the thermal conductivity of the specific surface area between the PCM and the fins is enhanced, while the reduced fin spacing inhibits natural convection, which also results in a much shorter melting time and a delayed solidification process for the PCM.

However, the heat transfer mechanism in the melting process of PCM embedded in metal foam is not dominated by convective heat transfer, but by heat conduction. This is because from the micro-scale heat transfer, it is concluded that the presence of foam metal skeleton acts as the heat transfer channel of phase change materials, so heat conduction is the main heat transfer mechanism in the melting process of PCM [34]. At the same time, the attenuation of convective heat transfer in liquid phase PCM is also an important reason due to the limitation of foam metal framework. Zilong Wang et al. [35] also used paraffin wax as the phase change material and added copper foam to it to prepare composite phase change materials with copper foam ratio increasing from 0 % to 2.13 %, respectively, and by studying the melting process of copper foam composite phase change materials, it was found that as the CMF ratio increased from 0.43 % to 2.13 %, the natural convection ratio decreased from 62.1 % to 7.7 % (results shown in Fig. 2), indicating that convective heat transfer is the main heat transfer mechanism in the melting process of low copper foam ratio composite phase change materials and heat conduction is the main heat transfer mechanism in the melting process of high copper foam ratio composite phase change materials.

For the numerical simulation part of the heat conduction and convective heat transfer mechanism of phase change materials, Mišo Jurčević et al. [36] developed for the first time a real-time numerical model capable of simulating higher scale solidification and melting in order to study the transient solidification of phase change materials dominated by convective heat transfer. Peiyi Li et al. [37] constructed a physical model of a slender rectangular cavity to numerically study the paraffin melting process, and also used several dimensionless parameters such as Reynolds number *Re* and Rayleigh number *Ra* as quantitative indicators to describe and evaluate the heat transfer mechanism in the cavity, and found that the convective heat transfer efficiency of PCM in the melting process can be achieved by increasing the Reynolds number *Re* and Rayleigh number *Ra*.

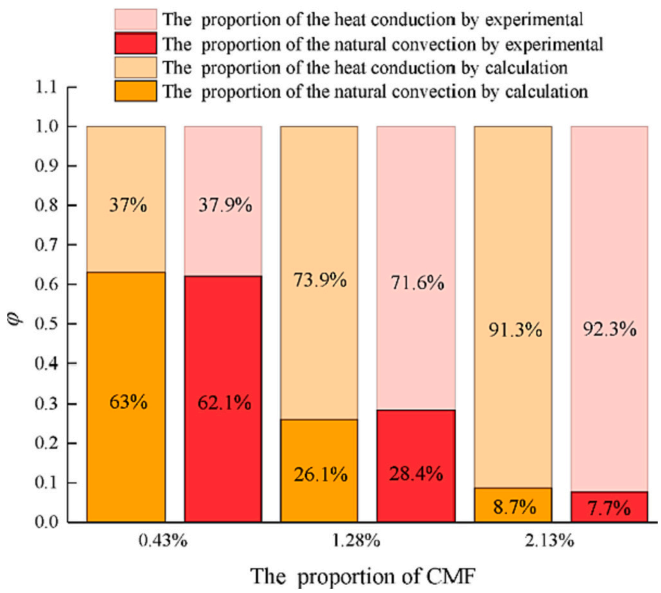

Fig. 2. Effect of the CMF proportion on the heat conduction and natural convection in the melting process of the composite PCMs [35].

2.1.2. Phonon heat transfer mechanism

Phonon conduction exists in the solid state of phase change materials, whether insulated or non-insulated, and phonons are the driving force responsible for transferring heat from one end to the other, so the study of phonon conduction mechanisms in micro-scale heat transfer is of great significance in controlling the thermal conductivity of phase change materials. The phonon heat transfer process is shown in Fig. 3. The atoms acquire thermal energy and give it to neighboring atoms in the form of vibrational waves, which then diffuse throughout the material and finally when transferred to the back of the material by conduction or radiation to the outside world.

Shaofei Wu et al. [38] reviewed the main influencing factors of phonon transport including phonon velocity, specific heat capacity, and mean free range. Particular attention should be paid to the fact that in non-metallic phase change materials, phonon scattering is the main cause of the thermal conductivity of the material, and phonon scattering causes the energy to be transferred in the opposite direction of the original. Phonon scattering can be divided into three parts: phonon-phonon scattering, phonon defect scattering and phonon interface scattering, as shown in Fig. 4. In general, the temperature dependence of the thermal conductivity depends on the specific heat capacity and is proportional to the absolute temperature (T). The mean free path of phonons refers to the average distance traveled by collisions between

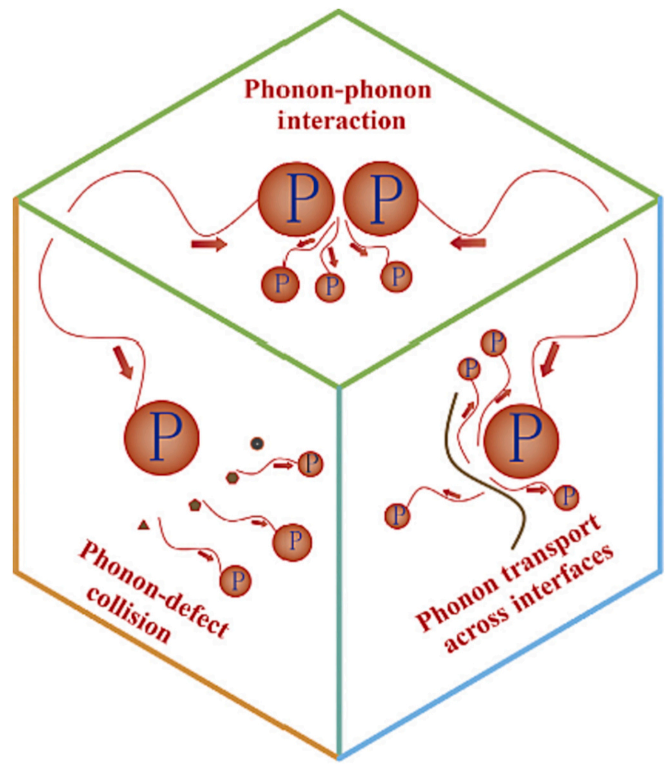

Fig. 4. Schematic diagram of three phonon scattering modes [38].

phonons. Generally, the higher the temperature, the more intense the collisions, the shorter the mean free path and the higher the thermal conductivity. For example, after grafting functional groups on the phase change material, the thermal resistance of the interface between the functional group and the phase change material decreases, thus the phonon scattering decreases, the average free range of phonons increases, and the thermal conductivity of the phase change material increases.

Fewer systematic studies analyze the thermal conductivity of phase change materials from the microscopic perspective of phonon heat conduction, but many researchers inevitably refer to phonon heat conduction mechanisms when studying thermal conductivity. For example, Haoxiang Lyu et al. [39] improved the thermal conductivity of the composite by 245 % when the graphene nanosheets (GNS) were doped at 3.0 wt%, due to a better phonon vibration match between GNS and sodium nitrate. C.Y. Zhao et al. [40] studied the thermal and phonon transport characteristics through MD simulation, and revealed that the weakening of phonon scattering in the nanolayer was the internal

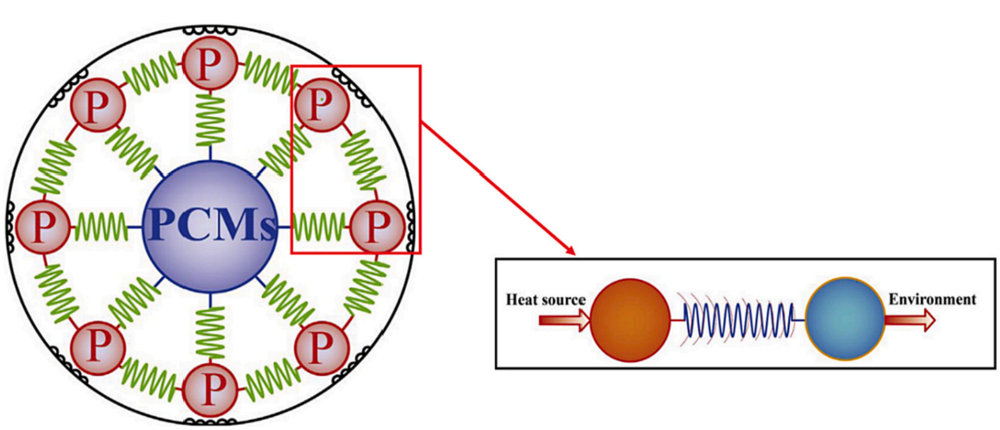

Fig. 3. Spring vibration heat transfer system composed of crystal atoms and abstract phonons [38].

mechanism for improving the thermal conductivity of NPCM.

2.1.3. Channel heat transfer mechanism

Organic phase change materials themselves carry functional groups, and these can be understood as heat transfer channels, and different functional groups have channels with different heat propagation rates. For example, multi-walled carbon nanotubes (MWCNT) contain hydroxyl (—OH), carboxyl (—COOH) and amino (—NH_2) groups, and the influence of functional groups on the phase transition behavior of composite PCM follows the following order: MWNT-COOH > MWNT-NH_2 > MWNT-OH > MWNT [41].

Not only the functional groups are part of the channel heat transfer mechanism, but also the interaction forces existing between the functional groups are part of the channel heat transfer mechanism. Hiroki Matsubara et al. [42] analyzed the molecular heat transfer mechanism of linear alcoholic liquids based on atomic heat path (AHP) analysis and showed that the hydroxyl group does not act as a heat carrier in the phase change material but provides the heat path for coulombic and intermolecular interactions, as shown in Fig. 5. Biao Feng et al. [43] analyzed the thermal conductivity of PE from the microscopic crystal structure of a solid-solid phase change material, pentaerythritol (PE), and verified by prediction and experiment that the main reason for the change in thermal conductivity during phase change is the breakage of strong intermolecular hydrogen bonds (HBs) on the crystal structure due to the increase in temperature and the formation of weak intermolecular HBs.

Therefore, one of the most common methods in improving the thermal conductivity of phase change materials is to increase the heat transfer channels, such as adding nanoparticles and embedding foam structures in PCM. For example, Zengkai Jiao et al. [44] increased the thermal conductivity by using nickel particles as catalysts to grow carbon nanotubes vertically onto a diamond film, treating the diamond skeleton as an "active" heat transfer and the carbon nanotubes as a "capillary" secondary heat transfer, and adding thermal conductivity channels to the phase change material.

2.2. Prediction model for thermal conductivity of phase change materials

In recent years, the most common methods to improve the thermal conductivity of phase change materials are the addition of nanomaterials (EPCM) to PCM and the filling of PCM into porous foams [44,45]. The thermal conductivity prediction models of these two methods are not quite the same, so the prediction models developed by domestic and foreign researchers are presented below for each of these two cases.

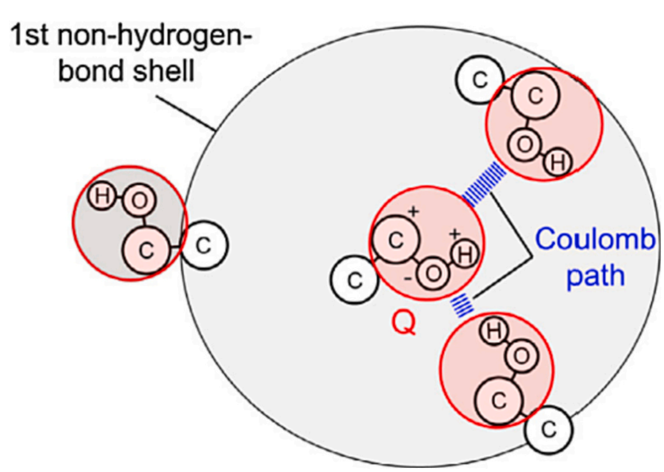

Fig. 5. The model of Coulomb path for ethanol as an example. The charge neutral group Q at the center has a Coulomb path with another Q inside the first nonhydrogen-bond shell. The ± symbols of the constituent atoms of Q at the center show the sign of partial electric charge [42].

2.2.1. Nanoparticle type

The earliest EPCM thermal conductivity prediction model is generally accepted in academia as the Maxwell model [46], but this model is only applicable to phase change materials where the nanoparticles are spherical. Later EVERY [47] changed the particle shape of the packing to derive a theoretical model of thermal conductivity applicable to ellipsoidal packing. Domestic and international thermal conductivity prediction models based on Maxwell's equation for nanofluid particle shapes are shown in Table 1.

In addition to developing various thermal conductivity prediction models for nanoparticle shapes, researchers have continued to refine thermal conductivity prediction models considering the nanofluid motion, viscosity, temperature, and particle volume concentration. Koo et al. [54] combined the Maxwell thermal conductivity prediction model with the Brownian motion of nanofluids thereby developing a new thermal conductivity prediction model that takes into account particle size, particle volume fraction and temperature dependence as well as the nature of the base fluid and particle phases with the following equations:

$$k_{eff} = k_f\left[\frac{k_p + 2k_f - 2\left(k_f - k_p\right)\phi}{k_p + 2k_f + \left(k_f - k_p\right)\phi}\right] + 5\times 10^4\beta\phi\rho_f C_{pf}\sqrt{\frac{kT}{\rho_p d_p}}f(T,\phi) \tag{1}$$

where $f(T,\phi)$ was fitted by Vajjha et al. [55] using a curve to produce a single correlation:

$$\mathrm{f}(\mathrm{T},\phi) = \left(2.8271\times 10^2\phi + 3.917\times 10^{-3}\right)\frac{T}{T_0} + \left(-3.0669\times 10^{-2}\phi - 3.91123\times 10^{-3}\right) \tag{2}$$

Also the thermal conductivity in nanofluidic composite phase change materials is inevitably affected by the interfacial thermal resistance [56], so several of the above predicted thermal conductivity models have some errors with the actual experimental values. To address the effect of interfacial thermal resistance, researchers have developed the Maxwell-Garnett type effective medium approach (EMA) [56], the improved Hamilton-Crosser model [57] and the Leong model [58],with the equations shown in Table 2.

Yanlin Song et al. [59] studied the prediction model of thermal conductivity of PCM composite of fatty acid and carbon materials based

Table 1
Thermal conductivity prediction model based on nanoparticle shape.

Model name	Particle shape	Model
Maxwell [46]	Spherical	$k_{\mathrm{eff}} = k_f\frac{k_p + 2k_f + 2\phi\left(k_p - k_f\right)}{k_p + 2k_f - \phi\left(k_p - k_f\right)}$
Bruggeman [48]	Spherical	$k_{\mathrm{eff}} = \frac{1}{4}\left[(3\phi - 1)k_p + (2 - 3\phi)k_f\right] + \frac{k_f}{4}\sqrt{\Delta}$ $\Delta = (3\phi - 1)^2(k_p/k_f)^2 + (2 - 3\phi)^2 + 2(2 + 9\phi - 9\phi^2)(k_p/k_f)$
Bhattacharya [49]	Spherical	$k_{eff} = \phi k_p + (1 - \phi)k_f$
Timofeeva [50]	Spherical	$k_{eff} = k_f(1 + 3\phi)$
Xue [51]	Nanotubes	$k_{eff} = k_f\frac{1 - \phi + 2\phi\frac{k_p}{k_p - k_f}ln\frac{k_p + k_f}{2k_f}}{1 - \phi + 2\phi\frac{k_f}{k_p - k_f}ln\frac{k_p + k_f}{2k_f}}$
Yamada-Ota [52]	Nanotubes	$k_{eff} = k_f\frac{1 + \frac{Hk_f}{rk_p}\phi^{0.2} + \left(1 - \frac{k_f}{k_p}\right)\phi\frac{H}{r}\phi^{0.2}}{1 + \frac{Hk_f}{rk_p}\phi^{0.2} - \left(1 - \frac{k_f}{k_p}\right)\phi}$
Yang [53]	Nanorods	$k_{eff} = \frac{(H + 2t)k_{\mathrm{eff}-x} + (r + t)k_{\mathrm{eff}-z}}{H + r + 3t}$

$k_{eff\text{-}x}$: Effective thermal conductivity on the x-axis; $k_{eff\text{-}z}$: Effective thermal conductivity in the z-axis; k_p: Thermal conductivity of nanoparticles; k_f: Thermal conductivity of the base fluid.

Table 2
Thermal conductivity prediction model for solving interface thermal resistance.

Model name	Model	Remarks
Nan [56]	$k_{eff} = k_f\left[\frac{3+\phi[2\beta_{xx}(1-L_{xx})+\beta_{zx}(1-L_{xz})]}{3-\phi[2\beta_{xx}(L_{xx})+\beta_{zz}(L_z)]}\right]$	$L_{xx} = \frac{a^2}{2(a^2-1)} + \frac{a}{2(1-a^2)^{3/2}}\cos^{-1}a$ $L_{\sum z} = (1-2L_{xx})$ $\beta_{xx} = \frac{k_{p,xx}-k_f}{k_f+L_{xx}(k_{p,xx}-k_f)}$ $k_{p,xx} = \frac{d_p}{2R_{eff}+d_p/k_{p,in-plane}}$ $a_k = R_k k_f$
Yu [57]	$k_{eff} = k_f\left[1+\frac{\phi a}{3}\frac{k_p/k_f}{a+\frac{2a_k}{\delta}\times\frac{k_p}{k_f}}\right]$	
Leong [58]	$k_{eff} = \frac{(k_p-k_{layer})\phi k_{layer}(2\beta_1{}^3-\beta_2{}^3+1)}{\beta_1{}^3(k_p+2k_{layer})-(k_p-k_{layer})\phi(\beta_1{}^3+\beta_2{}^3-1)} + \frac{(k_p+2k_{layer})\beta_1{}^3(\phi\beta_2{}^3(k_{layer}-k_f)+k_f)}{\beta_1{}^3(k_p+2k_{layer})-(k_p-k_{layer})\phi(\beta_1{}^3+\beta_2{}^3-1)}$	$\beta = h/r$ $\beta_1 = 1+\beta/2$ $\beta_2 = 1+\beta$ $k_{layer} = 10k_f$

a:Nanoparticle Aspect Ratio; R_{eff}: Effective interface thermal resistance; $k_{p,xx}$:modified in-plane thermal conductivity of Nanoparticle ; $k_{p,in\text{-}plane}$: In-plane thermal conductivity of Nanoparticle; k_{layer}: Nanosphere thermal conductivity; h: Nanosphere thickness.

on the Sierpinski carpet model with fractal geometry, and found that the QP prediction model and Maxwell-Eucken prediction model are more accurate, as shown in Fig. 6.

In recent years, not only special mathematical models have been developed to predict the thermal conductivity of different phase change materials for their characteristics [59], but also some machine learning methods have been used to do so. Jaliliantabar et al. [60] mainly introduced three machine learning methods to predict the thermal conductivity of nano-enhanced phase change materials, namely MARS (multiple adaptive regression splines), CART (classification and regression tree) and ANN (artificial neural network). Compared with past mathematical models for predicting the thermal conductivity of phase change materials, the models of machine learning methods are more general and avoid some complicated computational processes. However, one inevitable drawback of machine learning methods is that they require a sufficient amount of data to support them, and the quality of the data needs to be guaranteed.

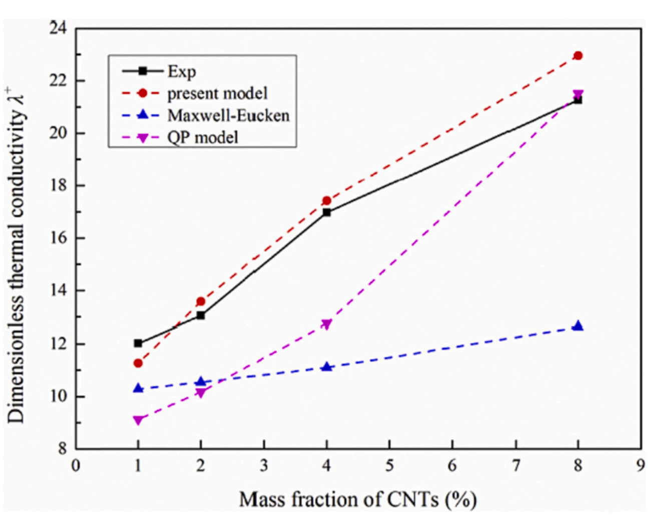

Fig. 6. Comparison of fractal prediction model of CNTs with QP and Maxwell-Eucken prediction model [59].

2.2.2. Metal foam type

From Section 2.1.3, we know that the thermal conductivity mechanism of filling the phase change material into the foam is to build heat transfer channels, and the more channel networks, the faster the heat transfer rate, the higher the thermal conductivity of this composite phase change material. Therefore, when predicting the use of metal foam to increase the thermal conductivity of phase change materials, one should understand the microscopic geometric model of metal foam as well as its ligament shape, orientation, vacuum, etc. The microscopic structure of metal foam is shown in Fig. 7.

For predicting the thermal conductivity of metal foam type composite phase change materials, researchers initially went on to derive a two-dimensional thermal conductivity prediction model based on the hexagonal structure of the foam metal matrix [62]. Later, Boomsma et al. [63] developed a three-dimensional thermal conductivity prediction model based on the structure of a foam matrix as a decahedron. However, the Boomsma-Poulikakos model ignores the metal foam ligament orientation factor when calculating the effective thermal conductivity, so the calculated thermal conductivity has some errors. Dai [64]. corrected and extended the Boomsma-Poulikakos model to take into account the metal foam ligament orientation factor to make the effective thermal conductivity prediction more accurate, as shown in Table 3. Dongxu Wu et al. [61] developed a model for predicting the effective thermal conductivity of open-cell foam composites based on the orthopentagonal dodecahedral geometry model.

The effective thermal conductivity obtained using thermal conductivity prediction models inevitably has some error with the values from experimental studies, and there are two general methods to assess the accuracy of various foam-metal-based thermal conductivity prediction models, namely the mean squared error (MSE) method [65] and the relative deviation (RD) method [66]. MSE is defined as

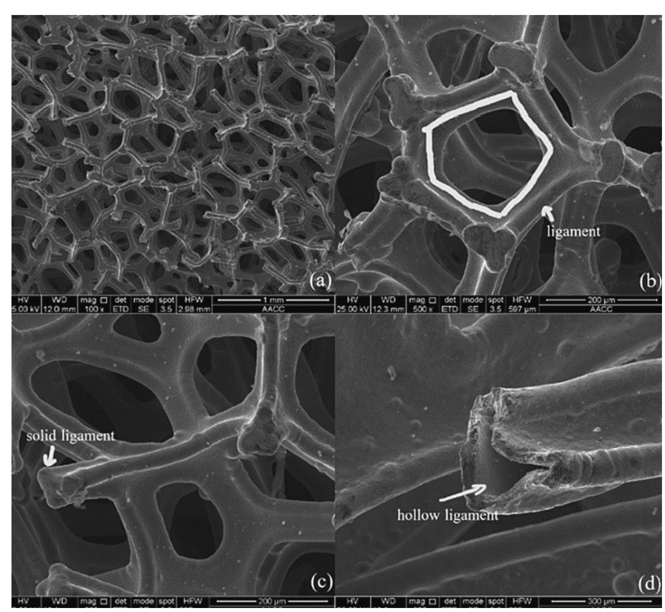

Fig. 7. Microstructure of open-cell foam: (a) the microstructure of the copper open cell foam; (b) typical unit cell structure of the open cell foam; (c) tri-prism solid ligaments; (d) hollow ligaments [61].

Table 3
Thermal conductivity prediction model based on foam metal substrate shape.

Model name	Substrate shape	Model	Remarks
Boomsma [63]	Decahedron	$k_{eff} = \frac{\sqrt{2}}{2(R_A + R_B + R_C + R_D)}$	$R_A = \frac{4g}{(2e^2 + \pi g(1-e))k_s + (4 - 2e^2 - \pi g(1-e))k_f}$ $R_B = \frac{(e-2g)^2}{(e-2g)e^2k_s + (2e - 4g - (e-2g)e^2)k_f}$ $R_C = \frac{(\sqrt{2}-2e)^2}{2\pi g^2(1-2e\sqrt{2})k_s + 2(\sqrt{2} - 2e - \pi g^2(1-2e\sqrt{2}))k_f}$ $R_D = \frac{2e}{e^2k_s + (4-e^2)k_f}$
Dai [64]	Decahedron	$k_{eff} = \frac{L_A + L_B + L_c + L_D}{R'_A + R'_B + R'_C + R'_D}$	$L_A + L_B + L_c + L_D = L\sqrt{2}/2$ $R_A' = LR_A$ $R_B' = LR_B$ $R_C' = LR_C$ $R'_D = LR_D$
Wu [61]	Orthopentagonal dodecahedron	$k_{eff} = \frac{L_H}{\left(\frac{2L_1}{k_1}\right) + \left(\frac{2L_2}{k_2}\right) + \left(\frac{2L_3}{k_3}\right) + \left(\frac{L_4}{k_4}\right)}$	$k_1 = 3.04\left(\frac{1+a_1^2}{a_1^2}\right)[2\varphi\theta + (1-4\varphi)\theta^2]k_s + \left\{1 - 3.04\left(\frac{1+a_1^2}{a_1^2}\right)[2\varphi\theta + (1-4\varphi)\theta^2]\right\}k_f$ $k_2 = 2.97\left(\frac{1+a_1^2}{a_1^2}\right)\varphi\theta^2k_s + \left(1 - 4.10\left(\frac{1+a_1^2}{a_1^2}\right)\varphi\theta^2\right)k_f$ $k_3 = 1.16\left(\frac{1+a_1^2}{a_1^2}\right)\theta^2k_s + \left(1 - 1.16\left(\frac{1+a_1^2}{a_1^2}\right)\theta^2\right)k_f$ $k_4 = 2.08\left(\frac{1+a_1^2}{a_1^2}\right)\varphi\theta^2k_s + \left(1 - 7.52\left(\frac{1+a_1^2}{a_1^2}\right)\varphi\theta^2\right)k_f$

k_s: Solid phase thermal conductivity of PCM ; k_f: Liquid phase thermal conductivity of PCM ; L_H: Height of unit pentagonal dodecahedron; R_n: Simplified quantities ; L_n: height of each layer ; n = A,B,C,D: unit cell subsection.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(k_{eff,exp} - k_{eff,pre}\right)^2 \tag{3}$$

and the relative deviation (RD) is calculated as

$$\mathrm{RD} = \left|\frac{k_{eff,pre} - k_{eff,exp}}{k_{eff,exp}}\right| \tag{4}$$

In addition to the above-mentioned prediction model of effective thermal conductivity derived by analyzing metal foam pores, ligaments, and thermal conductivity of filled phase change materials, Yao et al. [66] also proposed a new prediction model of effective thermal conductivity of high porosity open-cell metal foam by volume averaging and Fourier's heat transfer to law, and validated their model with experimental data of copper foam saturated with water, air, and paraffin.

3. Methods for improving the thermal conductivity of phase change materials

3.1. Nanoparticle dispersion into phase change materials

Recently, the application of nanocomposite phase change materials (NPCM) in energy storage systems has emerged as a promising approach [67], because NPCM meets our practical applications in various fields more than pure PCM [68–72]. The thermal conductivity of NPCM is higher than that of pure PCM, partly because the thermal conductivity of the nanomaterial itself is also much higher than that of PCM, as shown in Table 4. There are many types of nanomaterials, such as carbon-based nanomaterials, metal-based nanomaterials, etc. Carbon-based nanomaterials have stronger thermal conductivity compared to metal-based nanomaterials [73].

Nanoparticles can be divided into organic nanoparticles and inorganic nanoparticles. The addition of inorganic nanoparticles has a better effect on the thermophysical properties of phase change materials compared to organic nanoparticles. In contrast, organic nanoparticles are non-corrosive and therefore may be the best alternative for adding to corrosive salt hydrates [74].

From Section 2.1.2, we can know the microscopic mechanism of NPCM - the nanosphere induced by nanoparticles reduces the phonon scattering and thus promotes heat transfer, and the effect of NPCM is enhanced with the increase of nanoparticle mass fraction and temperature [40]. However, the increase in the thermal conductivity of PCM decreases as the concentration of nanoparticles gradually increases to a certain level due to the agglomeration effect that occurs when the number of nanoparticles is too large, resulting in a decrease in dispersion. The addition of appropriate surfactants (SAA) to PCM can improve these problems [85].

Table 4
Thermal Conductivities of some commonly used PCM and nanoparticle.

Materials	Thermal conductivity(W/m.K)	Reference
Phase change materials		
RT25(Paraffin wax)	0.25	[75]
Sebacic acid	0.26	[76]
D-Mannitol	1.308	[77]
Octadecane	0.17	[78]
Erythritol	0.73	[79]
Nanoparticles		
Silver nanoparticle (Ag)	429	[80]
Copper nanoparticle (Cu)	400	[81]
Titanium dioxide (TiO_2)	8.4	[81]
Expanded graphite (EG)	200	[82]
Carbon nanotube (CNT)	3000	[83]
Graphene	3000	[84]

3.1.1. Carbon based nanomaterials

Typical carbon-based nanomaterials (CBM) carbon nanotubes, graphite, and graphene have thermal conductivity almost five times higher than metals, and with the development of emerging technologies carbon derivatives have become the first choice for enhancing the thermal conductivity of PCM [86]. The magnitude of the increase in thermal conductivity by adding different CBMs to different PCMs is different, as shown in Table 5. CBM can be divided into one-dimensional (1D), two-dimensional (2D) and three-dimensional (3D) carbon-based materials, as shown in Fig. 8. 1D CBM can be better embedded in phase change materials. The thermal resistance between 2D CBM and

Table 5
Thermal conductivity enhancement of PCMs by carbon-based additives.

PCM	Thermal conductivity of PCM (W/m K)	Carbon additives	Fraction	The thermal conductivity of CPCM(W/m K)	Enhancement in thermal conductivity (%)	Reference
RT25(Paraffin wax)	0.260	CBNP	1 wt%	0.323	24.2	[75]
Soy Wax	0.324	Carbon nanofiber & CNTs	10 wt%	0.469 & 0.403	44.8 & 24.4	[88]
OAD(Octadecylamine)	0.330	GF	15 wt%	0.725	120	[89]
Palmitic acid matrix	0.220	MWCNT	1 wt%	0.330	50	[90]
$CaCl_2.6H_2O$	0.560	Reduced GO	0.18 wt %	1.008	80	[91]
$Na_2CO_3.10H_2O$/ $Na_2HPO_4.12H_2O$ (40:60)	0.688	Hydrothermal carbon (HTC)	1.5 wt%	0.990	44	[92]
Paraffin wax	0.29	rGOS	25 mg/ ml	0.87	200	[93]
Paraffin	0.260	EG	10 wt%	3.999	1438	[94]
Paraffin wax	0.155	CBNP	2.5 wt%	0.364	135	[95]

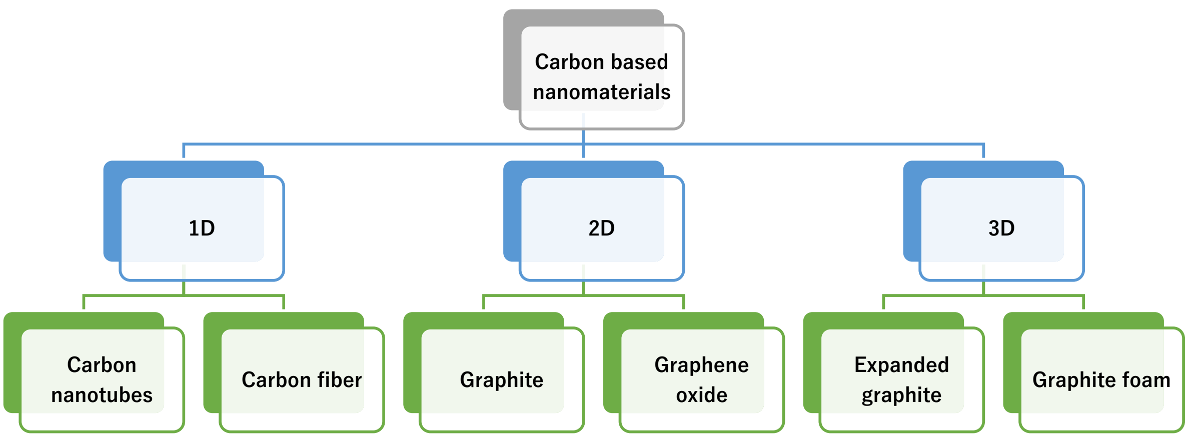

Fig. 8. Carbon based nanomaterials classification diagram.

PCM is smaller than 1D CBM and 3D CBM. The contact area between 3D CBM and CPM is increased, and 3D CBM is more suitable for PCM packaging [87].

If the nanomaterials are to be maximized, the key is the amount of 1D, 2D and 3D dimensional nanomaterials being added to the PCM. Alex Stonehouse et al. [96] found that the thermal conductivity of 3D carbon nano phase change materials can be enhanced by >300 % at low mass fractions (<5 %) and that 3D carbon nanomaterials show a positive linear relationship between mass fraction and TCE. Among the composites with high quality components, 1D and 2D have the highest thermal conductivity.

Carbon nanotubes (CNTs) are typical 1D CBMs, and the two most applied types of carbon nanotubes in existing research are single-walled carbon nanotubes (SWCNT) and multi-walled carbon nanotubes (MWCNT). From (a) and (b) of Fig. 9, it can be seen that the microstructures of SWCNT and MWCNT are different, so there is a difference in the extent to which adding to the same PCM improves their thermal conductivity. For example, when SWCNT and MWCNT were added to salt PCM, they were able to improve the maximum thermal conductivity of PCM by 56.98 % and 50.05 %, respectively [97].

Graphene (*Gr*) is a 2D structure composed of sp2 hybridized carbon atoms, and the microstructure is shown in Fig. 9(c). *Gr* is only the thickness of one carbon atom and there is flexibility in the interconnected carbon atoms [98]. The thermal conductivity of *Gr*/paraffin composites is higher than that of CNT/paraffin composites with the same filler mass fraction because of the excellent thermal and phonon vibration conductivity of the 2D structure of *Gr*, e.g. CNT has a higher thermal resistance than *Gr* [99]. Parameters such as specific heat capacity, density and viscosity of *Gr* also have a relatively large impact on the thermal conductivity. Elsaid et al. [100] addressed the problems faced by graphene-based nanofluids (NF) and suggested that because of the lower specific heat capacity, higher density and viscosity based on *Gr*-based NF, the volume fraction must be well optimized in the preparation of this material to withstand the increase in density and viscosity

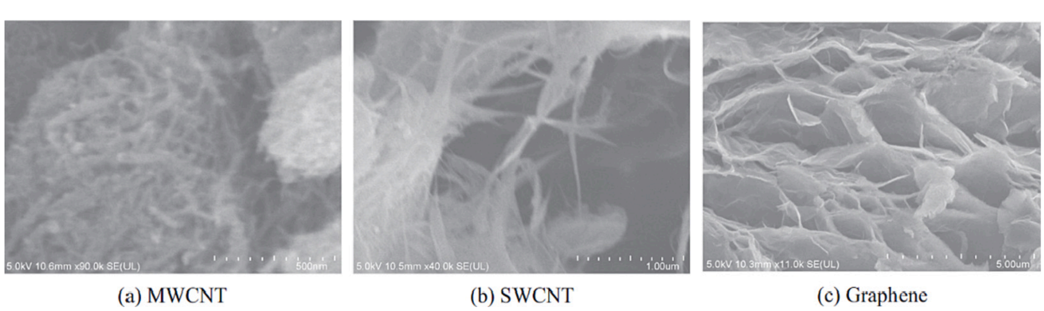

Fig. 9. Carbon nanomaterial microstructures by SEM [97].

to obtain a higher thermal conductivity.

Expanded graphite (MEG), a typical 3D CBM, is of great importance in improving the thermal conductivity of PCM. MEG is suitable for improving inorganic PCM as well as inorganic PCM. For example, Shaofei Wu et al. [101] added modified potassium aluminum sulfate dodecahydrate (MAPSD) as PCM to MEG and found that the composite samples with MEG addition had good heat transfer properties and higher thermal conductivity. The best thermal conductivity of the MEG/APSD composite for a sample density of 900 kg/m^3 and MEG content of 20 wt % is shown in Fig. 10. Yu et al. [102] studied paraffin/expanded graphite (PCM/MEG) composite phase change materials and found that the thermal conductivity of CM/MEG increased with the increase of MEG content and bulk density.

3.1.2. Metal based nanomaterials

Metal based nanomaterials can be broadly classified into two types of nanomaterials, pure metal and metal oxide, which improve the thermal conductivity of different PCMs, as shown in Table 6.

The thermal conductivity of the metal itself is very large, with silver (Ag) having the highest thermal conductivity (about 429 W/m.K), followed by copper (Cu). Also Ag has higher thermal conductivity than metal oxides, as Harikrishnan et al. [109] used 0.5 wt% of Al_2O_3, SiO_2, CuO, and Ag as nanomaterials and myristic acid as phase change material, and not only confirmed that the addition of metal based nanomaterials can enhance the thermal conductivity of myristic acid, but also Ag can make myristic acid obtain better performance, as shown in Fig. 11. Ag is also able to act with other metal oxide nanomaterials in PCM, and Dhivya et al. [110] significantly enhanced the thermal conductivity in oleic-myristic acid eutectic PCM microcapsules with high weight percentage of Ag-doped ZnO nanomaterials.

Although Ag has good thermal conductivity, it is relatively expensive and highly susceptible to oxidation in air, so domestic researchers usually choose other metals such as Cu as nanomaterials to improve thermal conductivity for PCMs that do not require too high thermal conductivity. For example, cut copper fibers can greatly improve the thermal conductivity of lithium-ion batteries using paraffin wax as PCM [111]. Xinghua Zheng et al. [112] studied the interfacial thermal conductivities (ITCs) of triplet metals (Cu, Ni, Al) and PCMs and found that Cu/PCMs < Ni/PCMs < Al/PCMs, which is due to the dominant role of lattice vibrations in the z-direction and the mismatch at the metal/PCMs interface, as shown in Fig. 12.

For metal oxide nanoparticles to enhance the thermal conductivity of PCM, Gupta et al. [113] added 0.5 wt% titanium dioxide (TiO_2), zinc oxide (ZnO), iron oxide (Fe_2O_3), and silicon dioxide (SiO_2), respectively,

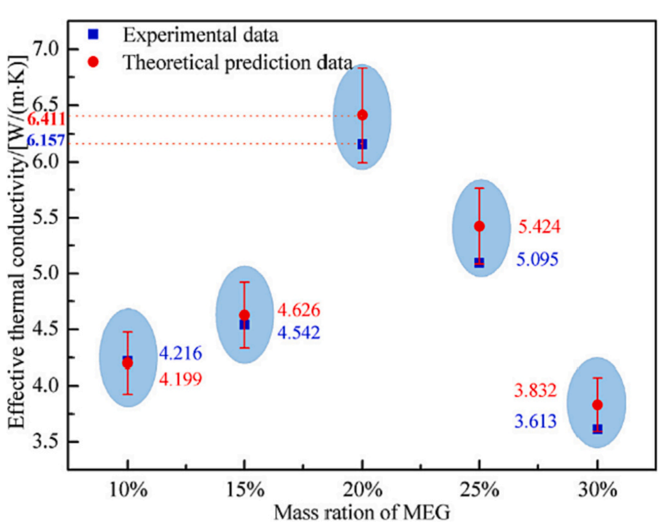

Fig. 10. Experimental and theoretical thermal conductivities of the MEG/APSD composite PCMs [101].

to magnesium nitrate hexahydrate (an inorganic salt hydrate PCM) and found that the addition of nanoparticles not only improved the thermal conductivity, the It also eliminated the subcooling phenomenon while maintaining the latent heat capacity. The analysis of the experimental results shows that the thermal conductivity of NPCM is increased by 147.5 %, 62.5 %, 55 % and 45 % by adding 0.5 wt% of TiO_2, ZnO, Fe_2O_3 and SiO_2, respectively.

3.2. Foam embedded phase change materials

Compared to the nanoparticles that can improve the thermal conductivity of PCM as described in Section 3.1, embedding the foam in PCM not only improves the thermal conductivity more strongly, but also has no stability problems and alleviates the generation of voids inside the PCM [114,115]. The improvement of thermal conductivity by foam embedded in PCM depends mainly on parameters such as foam material, foam porosity, pore size, and density of pores. This paper briefly discusses the more widely used metal and graphite foams.

3.2.1. Metal foam

Metal Foam (MF) as the name implies is that the foam material is metal. From Section 3.1.2, it can be seen that different metals have different thermal conductivity, so proper selection of metal foam is needed to improve the thermal conductivity of PCM. For example, Chaoming Wang et al. [116] found that the thermal conductivity could be increased by a maximum of 4.86 times when nickel foam was embedded in PCM compared to pure cetyl palmitate.

Among the parameters affecting the thermal conductivity of PCM-MFs the porosity and pore density of MF are the most important parameters [117]. Open-cell metal foams can be divided into two types: high porosity, with porosity above 85 %; low porosity, with porosity between 50 % and 85 % [118]. Regarding the porosity of MF, most national and international researchers have shown that lower porosity leads to higher thermal conductivity [114,119]. However, Hongyang Li et al. [120] found that MF inhibits the Rayleigh-B'enard convection of liquid PCM when the PCM is in the melting process of the phase transition, and the smaller the porosity of the MF, the stronger the inhibition of this convection, which affects the thermal conductivity of PCM-MFs, as also verified in Section 2.1.1 of this paper.

In general, the larger the pore density (ppi) of MF the weaker the natural convection and overall heat transfer of the composite phase change material, but recently Sheng Huang et al. [121] found that when the tilt angle is 90° and the Rayleigh number is 4.11E+7, the metal foam with higher pore density or smaller pore size has lower permeability, resulting in a 50ppi composite PCM natural convection is suppressed throughout the melting process, which reduces the heat flow to the top of the vessel. Although the total melting time is more than 10ppi, it is 6.25 % less than 25ppi melting time, as shown in Fig. 13.

3.2.2. Graphite foam

Graphite foam is a 3D high thermal conductivity material, and the microstructure is shown in Fig. 14. As early as 2008, researchers proposed to apply it to energy storage materials in space and on the ground [122], and in recent years it has been widely used in various fields [123–125]. Graphite foam is of great interest because of its excellent properties such as high thermal conductivity and very high melting point (4125 °C).

Heyao Zhang et al. [126] prepared graphite foam/erythritol phase change material and found that the thermal conductivity of the composite PCM increased from 40.52 W/(m.K) to 68.71 W/(m.K) and inhibited the supercooling of erythritol from 86.0 °C to 53.2 °C. Hao Lan et al. [127] also demonstrated that the thermal conductivity of graphite foam/PCM composite phase change material is >40 times that of pure chlorinated PCM. Similar to metal foams, the porosity of graphite foams can control the energy storage density and heat transfer rate of LHTES [128].

Table 6
Thermal conductivity enhancement of PCMs by metal additives.

PCM	Thermal conductivity of PCM (W/m K)	Metal additives	Fraction	The thermal conductivity of CPCM(W/m K)	Enhancement in Thermal Conductivity (%)	Reference
D-Mannitol	1.308	CuO	0.5 wt%	1.637	25.2	[77]
$Na_2SO_4.10H_2O–Na_2HPO_4.12H_2O$	0.789	α-Al_2O_3	4.55 %	1.278	62	[103]
PEG	0.297	Al_2O_3	3 wt%	0.435	46.5	[104]
MP-LA/PAN (100/100)	0.11	CNPs(Copper nanowires)	1.5 wt%	0.24	118	[105]
1-Tetradecanol	0.320	AgNW(Ag nanowires)	62.73 wt %	1.460	356	[106]
$BaCl_2.2H_2O$	0.6146	MgO	0.2 wt%	0.7215	17	[107]
Paraffin	0.25	Fe_3O_4	20 wt%	0.4	60	[108]

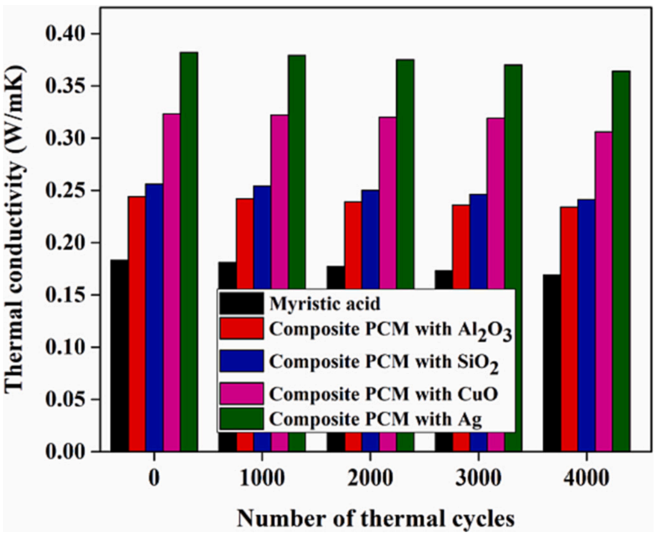

Fig. 11. Effect of thermal cycles on the thermal conductivity [109].

3.3. External field acting on phase change material

In addition to adding nanomaterials to PCM and embedding PCM in foam to improve thermal conductivity from the inside, it is also possible to obtain higher thermal conductivity by acting on PCM through external fields. In contrast to Sections 3.1 and 3.2, which focus on the enhancement of the heat transfer mechanism, the external field accelerates PCM melting by increasing the convective heat transfer mechanism [32,130,131], as described in Section 2.1.1. This paper summarizes three common external fields: ultrasonic, magnetic, and mechanical vibrations.

Ultrasonic vibration is accompanied by stirring, sound, flow, cavitation and oscillating fluid motion, which can effectively enhance conduction heat transfer and convection heat transfer [32], thus improving the thermal conductivity of PCM. Ultrasound can act directly on PCM to improve its thermal conductivity. For example, Nan Zhang et al. [132] used ultrasound to increase the heat transfer rate of the palmitate-stearate eutectic mixture (PA-SA) in a cylindrical stainless steel device. With the increase of ultrasonic power, the heat transfer rate of PA-SA smelting process increased, and when the ultrasonic power was 60 W, 105 W and 150 W, the heat transfer rate was increased by 10.64 %, 23.40 % and 31.91 %, respectively. With the increase of ultrasonic time, the heat transfer rate of PA-SA also increases. Ultrasonic waves can also indirectly act on the nanomaterials in PCM to improve the thermal conductivity of PCM. For example, Jeon et al. [133] found that the thermal conductivity of expanded graphite composite PCM prepared using ultrasonic waves was improved by about 40 % compared to the common stirring method, and both the dispersion stability and latent heat of the composite PCM were improved. Recently, Shang et al. [134] proposed that ultrasound can indirectly improve thermal conductivity by exfoliating graphite nanoflakes in composite PCM into nanoparticles.

The mechanism by which the magnetic field improves the thermal conductivity of PCM is through the control of flow and heat transfer by Lorentz and Kelvin forces [135]. For example, Kohyani et al. [136] investigated the phase change of cyclohexane-Cu nano-PCM in the presence of a magnetic field and found that the presence of a strong

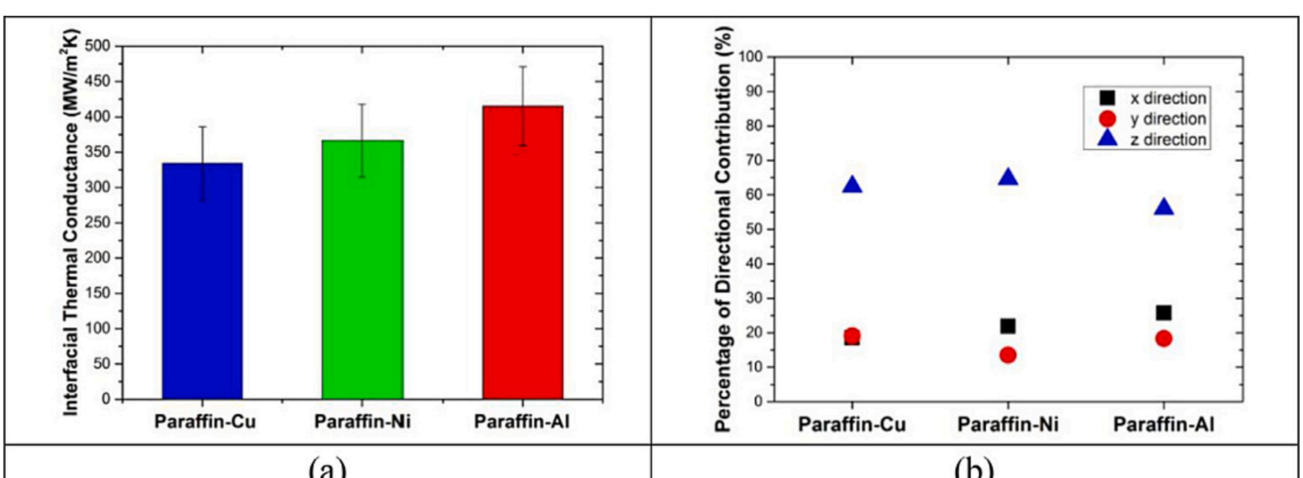

Fig. 12. (a) The ITCs of paraffin-Cu, paraffin-Ni and paraffin-Al model systems, (b) Relative contributions of lattice vibrations in x, y, and z directions to total heat flux for paraffin/metals model systems [112].

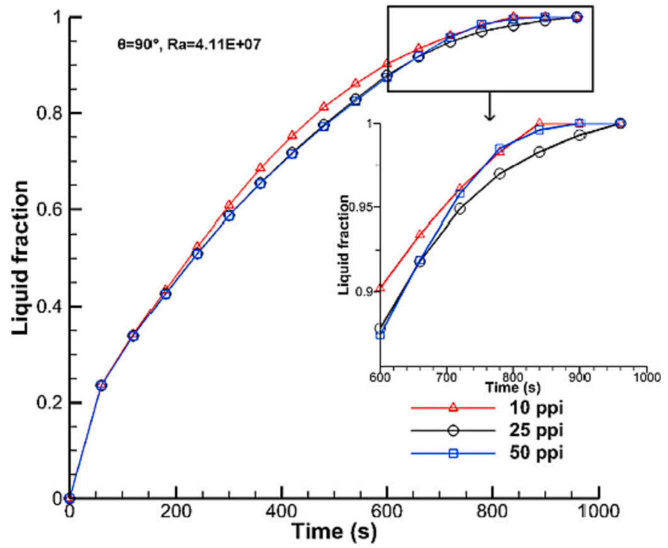

Fig. 13. Variations in liquid fraction throughout the melting process with different pore densities [121].

magnetic field accelerated the melting process and thus enhanced convective heat transfer. Yubin Fan et al. [130] investigated the circular motion of magnetic particles dispersed in liquid PCM in a heat source under the action of a rotating magnetic field and found that the rotating magnetic field led to forced convection in liquid PCM. The total melting time was reduced by 22.9 % when the magnetic field speed was 20 r/min, the particle fraction was 1 wt%, and the heating temperature was 35 °C.

Mechanical vibration can effectively control the crystal growth during the phase change process, thus improving the heat transfer characteristics of PCM during solidification. The effect of mechanical vibration on crystallization and heat transfer is a new field based on experimental and simulation studies. Zhang W et al. [137] conducted a series of experiments on the battery thermal management system of pure paraffin and three composite phase change materials (CPCMs) with 0–20 % mass fraction of expanded graphite or graphene, and found that proper mechanical vibration can enhance the heat transfer of CPCMs because mechanical vibration can accelerate the diffusion and collision of highly thermally conductive particles in CPCMs, and it was concluded that vibration has the best cooling effect on CPCMs with 20 % amount of expanded graphite, as shown in Fig. 15.

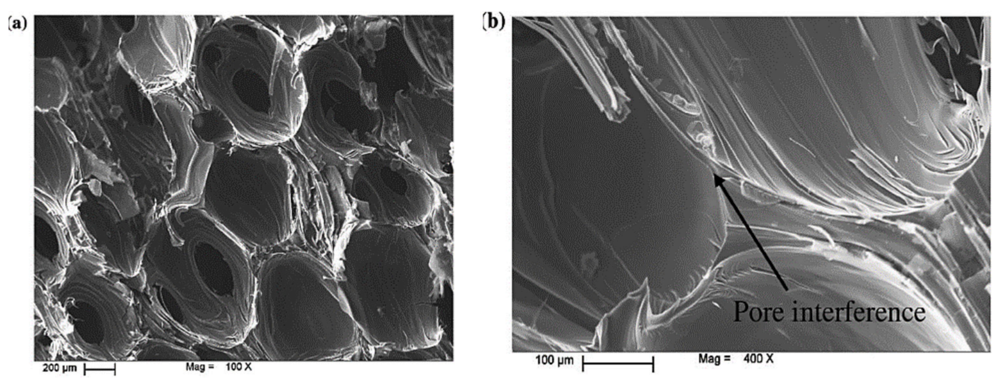

Fig. 14. Scanning electron microscope (SEM) images of the structure of graphite foam, showing the interconnected nearly-spherical pores of micro-structure in (a) 100×, and (b) 400× magnifications [129].

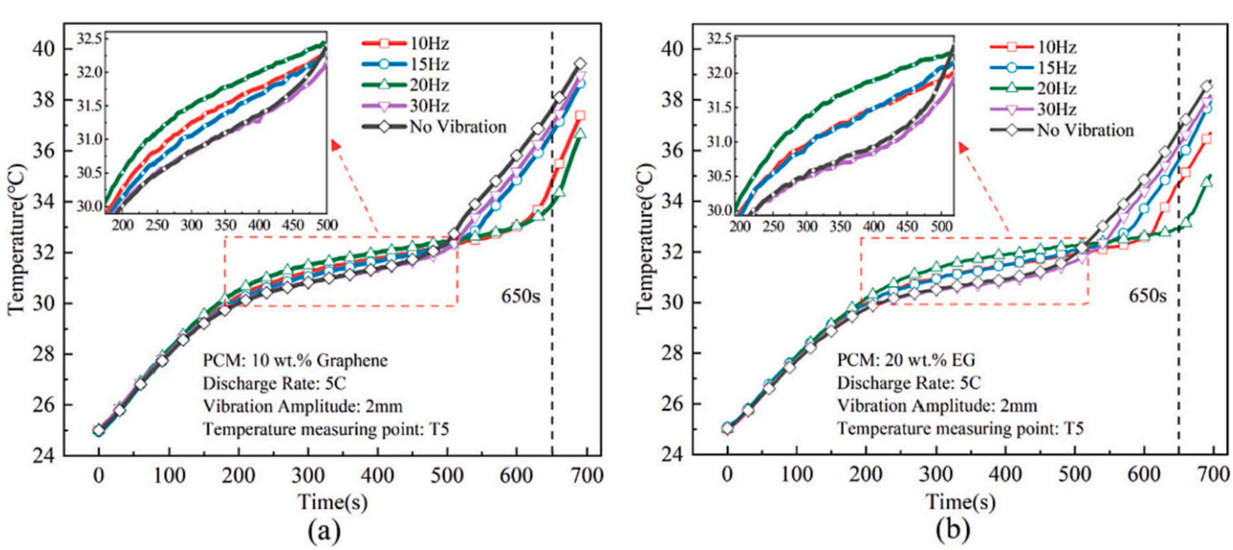

Fig. 15. Battery temperature at different vibration frequencies: (a) 10 wt% graphene with 90 wt% paraffin, 5C, 2 mm, T5, and (b) 20 wt% EG with 80 wt% paraffin, 5C, 2 mm, T5 [137].

4. Conclusion and outlook

In order to improve the thermal conductivity of PCM, this paper reviews the thermal conductivity mechanism, thermal conductivity prediction model and ways to improve the thermal conductivity of PCM from both microscopic and macroscopic perspectives. Some of the main conclusions can be drawn as follows:

(1) The basic heat transfer mechanism of phase change materials is reviewed, in which heat transfer can be subdivided into phonon heat transfer mechanism and channel heat transfer mechanism, which provides a theoretical basis for later experimental studies on improving the thermal conductivity of PCM.
(2) This paper reviews the thermal conductivity prediction models for nanoparticle-based and metal foam-based PCMs to provide data support for subsequent experimental studies to improve the thermal conductivity of PCMs upfront.
(3) Three specific ways to improve the thermal conductivity of PCM are reviewed and summarized, the addition of nanoparticles to PCM, the embedding of foam structures in PCM, and the addition of external fields in PCM.
(4) The internal mechanism of increasing the thermal conductivity of PCM by adding nanoparticles and embedded foam structure is to increase the heat transfer channels in PCM. The mechanism of increasing the thermal conductivity of PCM by adding external fields is to increase the convective heat transfer during the phase change of PCM.

The microscopic mechanisms and macroscopic ways to improve the thermal conductivity of PCM summarized and reviewed in this paper can provide the theoretical basis and experimental directions for future research on PCM, but there is still much research work to be done, such as:

(1) The thermal conduction mechanism of solid-gas and liquid-gas PCM is not clear. Both PCM have gaseous phase, so whether their thermal conduction mechanism is related to thermal radiation, and if so, what is the law of the thermal conduction mechanism.
(2) Is it possible to predict other foam materials, such as graphite foam, by correcting some coefficients of the foam metal type thermal conductivity prediction model.
(3) Nanoparticle-based thermal conductivity prediction model is developed based on the physical properties of nanoparticles added to PCM, similarly, can a thermal conductivity prediction model be developed for the combination of two PCMs (one PCM added to another PCM).
(4) Currently, there is a lack of predictive models for external fields to improve the thermal conductivity of PCM, and a new model can be developed from the power of ultrasound, the amplitude of vibration and other influencing factors.
(5) The external field is applied to the optimization of PCM method, and the laws of the four optimization modes of ultrasonic/magnetic field, ultrasonic/vibration, magnetic field/vibration and ultrasonic/magnetic field/vibration are compared and analyzed.
(6) The specific ways to improve the thermal conductivity of PCM are optimized, with a combination of two methods acting together on PCM, or even three ways together. At the same time, the law of their acting together on the thermal conductivity of PCM is explored.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Y. Wang, C.-H. Guo, C. Du, et al., Carbon peak and carbon neutrality in China: goals, implementation path, and prospects[J], China Geol. 4 (2021) 1–27.
[2] Y. Gao, X. Zhang, X. Xu, et al., Application and research progress of phase change energy storage in new energy utilization[J], J. Mol. Liq. 343 (2021).
[3] M.A. Keilany, S. Vannerem, M. Milhé, et al., Experimental and numerical study of combining encapsulated phase change material to sensible heat storage material in one-tank pilot scale thermal energy storage[J], J.Energy Storage (2022) 51.
[4] A.G. Fernández, L. González-Fernández, Y. Grosu, et al., Physicochemical characterization of phase change materials for industrial waste heat recovery applications[J], Energies 15 (10) (2022).
[5] Y. Zhao, C.Y. Zhao, C.N. Markides, et al., Medium- and high-temperature latent and thermochemical heat storage using metals and metallic compounds as heat storage media: a technical review[J], Appl. Energy 280 (2020).
[6] C. Ding, C. Zhang, L. Ma, et al., Numerical investigation on melting behaviour of phase change materials/metal foam composites under hypergravity conditions [J], Appl. Therm. Eng. 207 (2022).
[7] S. Ben Romdhane, A. Amamou, R. Ben Khalifa, et al., A review on thermal energy storage using phase change materials in passive building applications[J], <sb:contribution><sb:title>J. Build. </sb:title></sb:contribution><sb:host><sb:issue><sb:series><sb:title>Eng.</sb:title></sb:series></sb:issue></sb:host> (2020) 32.
[8] P. Royo, L. Acevedo, V.J. Ferreira, et al., High-temperature PCM-based thermal energy storage for industrial furnaces installed in energy-intensive industries[J], Energy 173 (2019) 1030–1040.
[9] A.F. Malaquias, S.F. Neves, J.B.L.M. Campos, Incorporation of phase change materials in fire protective clothing considering the presence of water[J], Int. J. Therm. Sci. 183 (2023).
[10] S. Rashidi, M. Samimifar, M.H. Doranehgard, et al., Organic phase change materials, in: Encyclopedia of Smart Materials, 2022, pp. 441–449.
[11] O. Moinuddin, G.V.N. Trivedi, R. Parameshwaran, et al., Preparation and characterization of microencapsulated organic phase change material for cool thermal energy storage applications[J], Mater.Today: Proc. 48 (2022) 639–640.
[12] A. Al-Ahmed, M.a.J. Mazumder, B. Salhi, et al., Effects of carbon-based fillers on thermal properties of fatty acids and their eutectics as phase change materials used for thermal energy storage: a review[J], J. Energy Storage (2021) 35.
[13] Z. Li, E. Gariboldi, Review on the temperature-dependent thermophysical properties of liquid paraffins and composite phase change materials with metallic porous structures[J], <sb:contribution><sb:title>Mater. Today</sb:title></sb:contribution><sb:host><sb:issue><sb:series><sb:title>Energy</sb:title></sb:series></sb:issue></sb:host> (2021) 20.
[14] S. Wang, Y. Xing, Z. Hao, et al., Experimental study on the thermal performance of PCMs based heat sink using higher alcohol/graphite foam[J], Appl. Therm. Eng. 198 (2021).
[15] S. Agarwala, K.N. Prabhu, Review of thermal characterization techniques for salt-based phase change materials[J], J.Energy Storage (2022) 46.
[16] M.F. Junaid, Z.U. Rehman, M. Čekon, et al., Inorganic phase change materials in thermal energy storage: a review on perspectives and technological advances in building applications[J], Energy Build. 252 (2021).
[17] Y. Lin, G. Alva, G. Fang, Review on thermal performances and applications of thermal energy storage systems with inorganic phase change materials[J], Energy 165 (2018) 685–708.
[18] S.A. Mohamed, F.A. Al-Sulaiman, N.I. Ibrahim, et al., A review on current status and challenges of inorganic phase change materials for thermal energy storage systems[J], Renew. Sust. Energ. Rev. 70 (2017) 1072–1089.
[19] I.A. Laghari, M. Samykano, A.K. Pandey, et al., Binary composite (TiO2-Gr) based nano-enhanced organic phase change material: effect on thermophysical properties[J], J.Energy Storage (2022) 51.
[20] Z. Sunxi, Z. Xuelai, L. Sheng, et al., Performance study on expand graphite/ organic composite phase change material for cold thermal energy storage[J], Energy Procedia 158 (2019) 5305–5310.
[21] X. Bao, H. Yang, X. Xu, et al., Development of a stable inorganic phase change material for thermal energy storage in buildings[J], Sol. Energy Mater. Sol. Cells 208 (2020).
[22] J. Hirschey, M. Goswami, D.O. Akamo, et al., Effect of expanded graphite on the thermal conductivity of sodium sulfate decahydrate (Na2SO4·10H2O) phase change composites[J], J.Energy Storage (2022) 52.
[23] S. Babu Sanker, R. Baby, Phase change material based thermal management of lithium ion batteries: a review on thermal performance of various thermal conductivity enhancers[J], J.Energy Storage (2022) 50.
[24] X. Lin, X. Zhang, L. Liu, et al., Polymer/expanded graphite-based flexible phase change material with high thermal conductivity for battery thermal management [J], J. Clean. Prod. 331 (2022).
[25] C. Ma, Y. Zhang, S. Hu, et al., A copper nanoparticle enhanced phase change material with high thermal conductivity and latent heat for battery thermal management[J], J. Loss Prev. Process Ind. 78 (2022).
[26] Y. Wang, Z. Wang, H. Min, et al., Performance investigation of a passive battery thermal management system applied with phase change material[J], J.Energy Storage 35 (2021).

[27] H. Liu, K. Sun, X. Shi, et al., Two-dimensional materials and their derivatives for high performance phase change materials: emerging trends and challenges[J], Energy Storage Mater. 42 (2021) 845–870.
[28] R. Anish, V. Mariappan, Joybari M. Mastani, Experimental investigation on the melting and solidification behavior of erythritol in a horizontal shell and multi-finned tube latent heat storage unit[J], Appl. Therm. Eng. (2019) 161.
[29] S. He, W. Wang, L. Wei, et al., Heat transfer enhancement and melting behavior of phase change material in a direct-contact thermal energy storage container[J], J. Energy Storage (2020) 31.
[30] M.A. Dekhil, J.V. Simo Tala, O. Bulliard-Sauret, et al., Numerical analysis of the performance enhancement of a latent heat storage shell and tube unit using finned tubes during melting and solidification[J], Appl. Therm. Eng. 192 (2021).
[31] X. Sun, Y. Chu, Y. Mo, et al., Experimental investigations on the heat transfer of melting phase change material (PCM)[J], Energy Procedia 152 (2018) 186.
[32] W. Cui, X. Li, X. Li, et al., Combined effects of nanoparticles and ultrasonic field on thermal energy storage performance of phase change materials with metal foam[J], Appl. Energy 309 (2022).
[33] X. Zhang, S. Wu, C. Zhang, et al., Dynamic heat transfer characteristics of gravity heat pipe with heat storage[J], J.Energy Storage (2022) 53.
[34] G. Wang, G. Wei, C. Xu, et al., Numerical simulation of effective thermal conductivity and pore-scale melting process of PCMs in foam metals[J], Appl. Therm. Eng. 147 (2019) 464–472.
[35] Z. Wang, H. Zhang, B. Dou, et al., Effect of copper metal foam proportion on heat transfer enhancement in the melting process of phase change materials[J], Appl. Therm. Eng. 201 (2022).
[36] M. Jurčević, Ž. Penga, B. Klarin, et al., Numerical analysis and experimental validation of heat transfer during solidification of phase change material in a large domain[J], J.Energy Storage (2020) 30.
[37] P. Li, Z. Wang, Z. Liu, et al., Convective heat transfer mechanisms of molten phase change material in a vertical slender rectangular cavity: a numerical case study [J], <sb:contribution><sb:title>Case Stud. </sb:title></sb:contribution><sb:host><sb:issue><sb:series><sb:title>Therm. Eng.</sb:title></sb:series></sb:issue></sb:host> (2021) 26.
[38] S. Wu, T. Yan, Z. Kuai, et al., Thermal conductivity enhancement on phase change materials for thermal energy storage: a review[J], Energy Storage Mater. 25 (2020) 251–295.
[39] H. Lyu, D. Feng, Y. Feng, et al., Enhanced thermal energy storage of sodium nitrate by graphene nanosheets: experimental study and mechanisms[J], J. Energy Storage (2022) 54.
[40] C.Y. Zhao, Y.B. Tao, Y.S. Yu, Molecular dynamics simulation of thermal and phonon transport characteristics of nanocomposite phase change material[J], J. Mol. Liq. 329 (2021).
[41] L. Feng, C. Wang, P. Song, et al., The form-stable phase change materials based on polyethylene glycol and functionalized carbon nanotubes for heat storage[J], Appl. Therm. Eng. 90 (2015) 952–956.
[42] H. Matsubara, G. Kikugawa, T. Bessho, et al., Molecular dynamics study on the role of hydroxyl groups in heat conduction in liquid alcohols[J], Int. J. Heat Mass Transf. 108 (2017) 749–759.
[43] B. Feng, J. Tu, J.-W. Sun, et al., A molecular dynamics study of the effects of crystalline structure transition on the thermal conductivity of pentaerythritol as a solid-solid phase change material[J], Int. J. Heat Mass Transf. 141 (2019) 789–798.
[44] Z. Jiao, S. Li, K. Zhou, et al., Application of multi-scale pore regulation for high thermal conductivity foam reinforcements in energy storage[J], Compos. Part A: Appl. Sci. Manuf. (2022) 157.
[45] M.a.M. Irwan, C.S.N. Azwadi, Y. Asako, et al., Review on numerical simulations for nano-enhanced phase change material (NEPCM) phase change process[J], J. Therm. Anal. Calorim. 141 (2) (2019) 669–684.
[46] J.C. Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., Clarendon Press, Oxford, UK, 1981 unabridged.
[47] A.G. Every, Y. Tzou, D. Hasselman, et al., in: The Effect of Particle Size on the Thermal Conductivity of ZnS/diamond Composites 40(1), 1992, pp. 123–129.
[48] D.J.a.O.P. Bruggeman, in: The Calculation of Various Physical Constants of Heterogeneous Substances. I. The Dielectric Constants and Conductivities of Mixtures Composed of Isotropic Substances, 1935, p. 416.
[49] P. Bhattacharya, S.K. Saha, A. Yadav, et al., Brownian dynamics simulation to determine the effective thermal conductivity of nanofluids[J], J. Appl. Phys. 95 (11) (2004) 6492–6494.
[50] E.V. Timofeeva, A.N. Gavrilov, J.M. Mccloskey, et al., Thermal conductivity and particle agglomeration in alumina nanofluids: experiment and theory[J], Phys. Rev. E Stat. Nonlinear Soft Matter Phys. 76 (6 Pt 1) (2007), 061203.
[51] Q.Z. Xue, Model for thermal conductivity of carbon nanotube-based composites [J], Phys. B Condens. Matter 368 (1–4) (2005) 302–307.
[52] E.J.N.B. Yamada, in: The Effective Thermal Conductivity of Dispersed Materials 3 (2), 1989, pp. 78–83.
[53] L. Yang, X. Xu, W. Jiang, et al., A new thermal conductivity model for nanorod-based nanofluids[J], Appl. Therm. Eng. 114 (2017) 287–299.
[54] J. Koo, C. Kleinstreuer, A new thermal conductivity model for nanofluids[J], J. Nanopart. Res. 6 (6) (2005) 577–588.
[55] R.S. Vajjha, D.K. Das, P.K. Namburu, Numerical study of fluid dynamic and heat transfer performance of Al2O3 and CuO nanofluids in the flat tubes of a radiator [J], Int. J. Heat Fluid Flow 31 (4) (2010) 613–621.
[56] C.-W. Nan, G. Liu, Y. Lin, et al., Interface effect on thermal conductivity of carbon nanotube composites[J], Appl. Phys. Lett. 85 (16) (2004) 3549–3551.
[57] W. Yu, S.U.S. Choi, The role of interfacial layers in the enhanced thermal conductivity of nanofluids: a renovated Hamilton-Crosser model[J], J. Nanopart. Res. 6 (4) (2004) 355–361.
[58] K.C. Leong, C. Yang, S.M.S. Murshed, A model for the thermal conductivity of nanofluids – the effect of interfacial layer[J], J. Nanopart. Res. 8 (2) (2006) 245–254.
[59] Y. Song, N. Zhang, Y. Yuan, et al., Prediction of the solid effective thermal conductivity of fatty acid/carbon material composite phase change materials based on fractal theory[J], Energy 170 (2019) 752–762.
[60] F. Jaliliantabar, Thermal conductivity prediction of nano enhanced phase change materials: a comparative machine learning approach[J], J.Energy Storage (2022) 46.
[61] D. Wu, C. Huang, Thermal conductivity model of open-cell foam suitable for wide span of porosities[J], Int. J. Heat Mass Transf. 130 (2019) 1075–1086.
[62] V.V. Calmidi, R.L. Mahajan, The effective thermal conductivity of high porosity fibrous metal foams[J], J. Heat Transf. 121 (2) (1999) 466–471.
[63] K. Boomsma, D. Poulikakos, On the effective thermal conductivity of a three-dimensionally structured fluid-saturated metal foam[J], Int. J. Heat Mass Transfer 44 (4) (2001) 827–836.
[64] Z. Dai, K. Nawaz, Y.G. Park, et al., Correcting and extending the Boomsma-Poulikakos effective thermal conductivity model for three-dimensional, fluid-saturated metal foams[J], Int.Commun.Heat Mass Transf. 37 (6) (2010) 575–580.
[65] P. Ranut, On the effective thermal conductivity of aluminum metal foams: review and improvement of the available empirical and analytical models[J], Appl. Therm. Eng. 101 (2016) 496–524.
[66] Y. Yao, H. Wu, Z. Liu, A new prediction model for the effective thermal conductivity of high porosity open-cell metal foams[J], Int. J. Therm. Sci. 97 (2015) 56–67.
[67] S. Nižetić, M. Jurčević, M. Arıcı, et al., Nano-enhanced phase change materials and fluids in energy applications: a review[J], Renew. Sust. Energ. Rev. 129 (2020).
[68] K.W. Shah, A review on enhancement of phase change materials - a nanomaterials perspective[J], Energy Build. 175 (2018) 57–68.
[69] A.S. Abdelrazik, R. Saidur, F.A. Al-Sulaiman, Thermal regulation and performance assessment of a hybrid photovoltaic/thermal system using different combinations of nano-enhanced phase change materials[J], Sol. Energy Mater. Sol. Cells 215 (2020).
[70] S.I. Ibrahim, A.H. Ali, S.A. Hafidh, et al., Stability and thermal conductivity of different nano-composite material prepared for thermal energy storage applications[J], S.Afr.J.Chem.Eng. 39 (2022) 72–89.
[71] M. Qin, A. Almarashi, Z.J. Talabany, et al., Charging of phase change material layers though air heat exchanger considering TiO2 nanomaterial[J], J.Energy Storage (2022) 47.
[72] S. Rostami, A.A. Nadooshan, A. Raisi, et al., Effect of using a heatsink with nanofluid flow and phase change material on thermal management of plate lithium-ion battery[J], J.Energy Storage (2022) 52.
[73] B. Eanest Jebasingh, Arasu A. Valan, A comprehensive review on latent heat and thermal conductivity of nanoparticle dispersed phase change material for low-temperature applications[J], Energy Storage Mater. 24 (2020) 52–74.
[74] L.-S. Wong-Pinto, Y. Milian, S. Ushak, Progress on use of nanoparticles in salt hydrates as phase change materials[J], Renew. Sust. Energ. Rev. 122 (2020).
[75] L. Colla, L. Fedele, S. Mancin, et al., Nano-PCMs for enhanced energy storage and passive cooling applications[J], Appl. Therm. Eng. 110 (2017) 584–589.
[76] Q. Zhang, J. Liu, Sebacic acid/CNT sponge phase change material with excellent thermal conductivity and photo-thermal performance[J], Sol. Energy Mater. Sol. Cells 179 (2018) 217–222.
[77] S. Salyan, S. Suresh, Study of thermo-physical properties and cycling stability of d -Mannitol-copper oxide nanocomposites as phase change materials[J], J.Energy Storage 15 (2018) 245–255.
[78] S. Motahar, N. Nikkam, A.A. Alemrajabi, et al., Experimental investigation on thermal and rheological properties of n-octadecane with dispersed TiO2 nanoparticles[J], Int. Commun. Heat Mass Transf. 59 (2014) 68–74.
[79] T. Nomura, K. Tabuchi, C. Zhu, et al., High thermal conductivity phase change composite with percolating carbon fiber network[J], Appl. Energy 154 (2015) 678–685.
[80] R.M. Al Ghossein, M.S. Hossain, J.M. Khodadadi, Experimental determination of temperature-dependent thermal conductivity of solid eicosane-based silver nanostructure-enhanced phase change materials for thermal energy storage[J], Int. J. Heat Mass Transf. 107 (2017) 697–711.
[81] A. Sciacovelli, F. Colella, V. Verda, Melting of PCM in a thermal energy storage unit: numerical investigation and effect of nanoparticle enhancement[J], Int. J. Energy Res. 37 (13) (2013) 1610–1623.
[82] T. Oya, T. Nomura, M. Tsubota, et al., Thermal conductivity enhancement of erythritol as PCM by using graphite and nickel particles[J], Appl. Therm. Eng. 61 (2) (2013) 825–828.
[83] L. Liu, D. Su, Y. Tang, et al., Thermal conductivity enhancement of phase change materials for thermal energy storage: a review[J], Renew. Sust. Energ. Rev. 62 (2016) 305–317.
[84] S.S. Park, N.J. Kim, Influence of the oxidation treatment and the average particle diameter of graphene for thermal conductivity enhancement[J], J. Ind. Eng. Chem. 20 (4) (2014) 1911–1915.
[85] S.S. Prabhu, P. Selvakumar, J.S. Heric, Thermal stability and behaviour of paraffin nano Al2O3, graphene and surfactant sodium oleate composite as phase change material[J], Mater.Res.Express 9 (11) (2022).
[86] S.S. Yao, F.L. Jin, K.Y. Rhee, et al., Recent Advances in Carbon-fiber-reinforced Thermoplastic Composites: A Review, 2018. S1359836817322692.

[87] A.G. Olabi, T. Wilberforce, K. Elsaid, et al., Recent progress on carbon-based nanomaterial for phase change materials: prospects and challenges[J], Therm.Sci. Eng.Prog. (2021) 23.

[88] Y. Cui, C. Liu, S. Hu, et al., The experimental exploration of carbon nanofiber and carbon nanotube additives on thermal behavior of phase change materials[J], Sol. Energy Mater. Sol. Cells 95 (4) (2011) 1208–1212.

[89] L. Jiang, L. Zhao, G. Chen, et al., Graphite-enhanced thermal properties of octadecylamine/graphite foam shape-stable composite phase change materials for thermal energy storage[J], Energy Rep. 8 (2022) 13939–13947.

[90] J. Wang, H. Xie, Z. Xin, et al., Enhancing thermal conductivity of palmitic acid based phase change materials with carbon nanotubes as fillers[J], Sol. Energy 84 (2) (2010) 339–344.

[91] X. Zhang, X. Li, Y. Zhou, et al., Enhanced thermal conductivity in a hydrated salt PCM system with reduced graphene oxide aqueous dispersion[J], RSC Adv. 8 (2) (2018) 1022–1029.

[92] Wang, Jie, , et al.Han, in: Hydrothermal Carbon Doped Form-Stable Inorganic Hydrate Salts Phase Change Materials With Excellent Reutilization 6(7), 2018, pp. 1220–1227.

[93] P. Li, Q. Chen, Q. Peng, et al., Paraffin/graphene sponge composite as a shape-stabilized phase change material for thermal energy storage[J], Pigm. Resin Technol. 50 (5) (2020) 412–418.

[94] L. Xia, P. Zhang, R.Z. Wang, Preparation and thermal characterization of expanded graphite/paraffin composite phase change material[J], Carbon 48 (9) (2010) 2538–2548.

[95] A.K. Mishra, B.B. Lahiri, J. Philip, Superior thermal conductivity and photo-thermal conversion efficiency of carbon black loaded organic phase change material[J], J. Mol. Liq. 285 (2019) 640–657.

[96] A. Stonehouse, C. Abeykoon, Thermal properties of phase change materials reinforced with multi-dimensional carbon nanomaterials[J], Int. J. Heat Mass Transf. 183 (2022).

[97] Y.B. Tao, C.H. Lin, Y.L. He, Preparation and thermal properties characterization of carbonate salt/carbon nanomaterial composite phase change material[J], Energy Convers. Manag. 97 (2015) 103–110.

[98] A. Yadav, A. Verma, A. Kumar, et al., Recent advances on enhanced thermal conduction in phase change materials using carbon nanomaterials[J], J.Energy Storage (2021) 43.

[99] X. Wu, M. Gao, K. Wang, et al., Experimental study of the thermal properties of a homogeneous dispersion system of a paraffin-based composite phase change materials[J], J.Energy Storage 36 (2021).

[100] K. Elsaid, M.A. Abdelkareem, H.M. Maghrabie, et al., Thermophysical properties of graphene-based nanofluids[J], Int.J.Thermofluids 10 (2021).

[101] S. Wu, T. Yan, Z. Kuai, et al., Experimental and numerical study of modified expanded graphite/hydrated salt phase change material for solar energy storage [J], Sol. Energy 205 (2020) 474–486.

[102] X.K. Yu, Y.B. Tao, Preparation and characterization of paraffin/expanded graphite composite phase change materials with high thermal conductivity[J], Int. J. Heat Mass Transf. 198 (2022).

[103] Y. Liu, Y. Yang, Use of nano-α-Al2O3 to improve binary eutectic hydrated salt as phase change material[J], Sol. Energy Mater. Sol. Cells 160 (2017) 18–25.

[104] B. Tang, C. Wu, M. Qiu, et al., PEG/SiO2–Al2O3 hybrid form-stable phase change materials with enhanced thermal conductivity[J], Mater. Chem. Phys. 144 (1–2) (2014) 162–167.

[105] N. Xie, J. Niu, X. Gao, et al., Fabrication and characterization of electrospun fatty acid form-stable phase change materials in the presence of copper nanoparticles [J], Int. J. Energy Res. 44 (11) (2020) 8567–8577.

[106] J.L. Zeng, Z. Cao, D.W. Yang, et al., Thermal conductivity enhancement of Ag nanowires on an organic phase change material[J], J. Therm. Anal. Calorim. 101 (1) (2009) 385–389.

[107] J.M. Munyalo, X. Zhang, X. Xu, Experimental investigation on supercooling, thermal conductivity and stability of nanofluid based composite phase change material[J], J.Energy Storage 17 (2018) 47–55.

[108] N. Şahan, M. Fois, H. Paksoy, Improving thermal conductivity phase change materials—a study of paraffin nanomagnetite composites[J], Sol. Energy Mater. Sol. Cells 137 (2015) 61–67.

[109] S. Harikrishnan, A.D. Dhass, H.F. Oztop, et al., Measurement of thermophysical properties with nanomaterials on the melting/freezing characteristics of phase change material[J], Measurement 199 (2022).

[110] S. Dhivya, S.I. Hussain, S. Jeya Sheela, et al., Experimental study on microcapsules of Ag doped ZnO nanomaterials enhanced oleic-myristic acid eutectic PCM for thermal energy storage[J], Thermochim. Acta 671 (2019) 70–82.

[111] M. Pan, W. Lai, Cutting copper fiber/paraffin composite phase change material discharging experimental study based on heat dissipation capability of Li-ion battery[J], Renew. Energy 114 (2017) 408–422.

[112] X. Zheng, X. Wang, T. Zhang, et al., Study on the interfacial thermal conductance between metals and phase change materials[J], Int. J. Heat Mass Transf. 168 (2021).

[113] N. Gupta, A. Kumar, H. Dhasmana, et al., Enhanced thermophysical properties of metal oxide nanoparticles embedded magnesium nitrate hexahydrate based nanocomposite for thermal energy storage applications[J], J.Energy Storage (2020) 32.

[114] M. Al-Jethelah, S. Ebadi, K. Venkateshwar, et al., Charging nanoparticle enhanced bio-based PCM in open cell metallic foams: an experimental investigation[J], Appl. Therm. Eng. 148 (2019) 1029–1042.

[115] Z.G. Wu, C.Y. Zhao, Experimental investigations of porous materials in high temperature thermal energy storage systems[J], Sol. Energy 85 (7) (2011) 1371–1380.

[116] C. Wang, T. Wang, Z. Hu, et al., Facile synthesis and thermal performance of cety palmitate/nickel foam composite phase change materials for thermal energy storage[J], J.Energy Storage (2020) 28.

[117] M. Aramesh, B. Shabani, Metal foam-phase change material composites for thermal energy storage: a review of performance parameters[J], Renew. Sust. Energ. Rev. 155 (2022).

[118] F. Zhu, X. Hu, X. Wang, et al., Experimental and numerical investigation of the melting process of aluminum foam/paraffin composite with low porosity[J], Numer.Heat Transf.Part A Appl. 77 (12) (2020) 998–1013.

[119] X. Hu, F. Zhu, X. Gong, Experimental and numerical study on the thermal behavior of phase change material infiltrated in low porosity metal foam[J], J. Energy Storage (2019) 26.

[120] H. Li, C. Hu, Y. He, et al., Pore-scale study on Rayleigh-Bénard convection formed in the melting process of metal foam composite phase change material[J], Int. J. Therm. Sci. 177 (2022).

[121] S. Huang, J. Lu, Y. Li, Numerical study on the influence of inclination angle on the melting behaviour of metal foam-PCM latent heat storage units[J], Energy 239 (2022).

[122] K. Lafdi, O. Mesalhy, A. Elgafy, Graphite foams infiltrated with phase change materials as alternative materials for space and terrestrial thermal energy storage applications[J], Carbon 46 (1) (2008) 159–168.

[123] P. Jana, E. Palomo Del Barrio, M. Dubois, et al., Hydrophobised carbon foams for improved long-term seasonal solar thermal energy storage[J], Sol. Energy Mater. Sol. Cells 220 (2021).

[124] R. Chen, R. Yao, W. Xia, et al., Electro/photo to heat conversion system based on polyurethane embedded graphite foam[J], Appl. Energy 152 (2015) 183–188.

[125] H. Badenhorst, N. Fox, A. Mutalib, The use of graphite foams for simultaneous collection and storage of concentrated solar energy[J], Carbon 99 (2016) 17–25.

[126] H. Zhang, J. Cheng, Q. Wang, et al., The graphite foam/erythritol composites with ultrahigh thermal conductivity for medium temperature applications[J], Sol. Energy Mater. Sol. Cells 230 (2021).

[127] H. Lan, S. Dutta, N. Vahedi, et al., Graphite foam infiltration with mixed chloride salts as PCM for high-temperature latent heat storage applications[J], Sol. Energy 209 (2020) 505–514.

[128] D. Singh, T. Kim, W. Zhao, et al., Development of graphite foam infiltrated with MgCl2 for a latent heat based thermal energy storage (LHTES) system[J], Renew. Energy 94 (2016) 660–667.

[129] M. Moeini Sedeh, J.M. Khodadadi, Thermal conductivity improvement of phase change materials/graphite foam composites[J], Carbon 60 (2013) 117–128.

[130] Y. Fan, M. Yu, C. Zhang, et al., Melting performance enhancement of phase change material with magnetic particles under rotating magnetic field[J], J. Energy Storage 38 (2021).

[131] X. Wang, R.-G. Guan, Heat transfer and cooling rate model of flow melt on vibration wall[J], Trans. Indian Inst. Metals 71 (7) (2018) 1635–1641.

[132] N. Zhang, Y. Du, Ultrasonic enhancement on heat transfer of palmitic-stearic acid as PCM in unit by experimental study[J], Sustain. Cities Soc. 43 (2018) 532–537.

[133] J. Jeon, J.H. Park, S. Wi, et al., Thermal performance enhancement of a phase change material with expanded graphite via ultrasonication[J], J. Ind. Eng. Chem. 79 (2019) 437–442.

[134] B. Shang, R. Wu, J. Hu, et al., Non-monotonously tuning thermal conductivity of graphite-nanosheets/paraffin composite by ultrasonic exfoliation[J], Int. J. Therm. Sci. 131 (2018) 20–26.

[135] A.E. Kabeel, E.M.S. El-Said, S.A. Dafea, A review of magnetic field effects on flow and heat transfer in liquids: present status and future potential for studies and applications[J], Renew. Sust. Energ. Rev. 45 (2015) 830–837.

[136] M.T. Kohyani, B. Ghasemi, A. Raisi, et al., Melting of cyclohexane–Cu nano-phase change material (nano-PCM) in porous medium under magnetic field[J], J. Taiwan Inst. Chem. Eng. 77 (2017) 142–151.

[137] W. Zhang, X. Li, W. Wu, et al., Influence of mechanical vibration on composite phase change material based thermal management system for lithium-ion battery [J], J.Energy Storage (2022) 54.